MATH 241 Spring 2026 Homework #9 (Last Homework for Exam 02)

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May 3, 2026

These exercises are designed to review Section 4.2.

What is the formula we use to compute $\mu = E[X]$?
$E[X] := \sum_{\text{all } x} x f_X(x)$. Can you intuitively/geometrically explain what is meant by $\mu = E[X]$? Geometrically, the expected value represents the balance point of the distribution. If the x-axis is imagined as a weightless rod with weights of size $f(x)$ placed at each x, $E[X]$ is the point where the rod balances. Suppose X is a discrete random variable with probability mass function f. What is the difference between

$$\sum_{\text{all } x} f_X(x) \quad \text{and} \quad \sum_{\text{all } x} x f_X(x)?$$

$\sum f_X(x) = 1$ (the sum of all probabilities in a pmf).
$\sum x f_X(x) = E[X]$ (the weighted average or expected value). Each of two coins are to be flipped exactly once. The first coin will land on heads with probability 0.6, and the second coin will land on heads with probability 0.7. Assume that the results of the flips are independent of each other and let X denote that results. Find $E[X]$.
Let X_1 and X_2 be Bernoulli trials for each coin.
$E[X_1] = 0.6$ and $E[X_2] = 0.7$.
Since $E[X + Y] = E[X] + E[Y]$, the total expected heads is $0.6 + 0.7 = \mathbf{1.3}$. What does The Law of The Unconscious Statistician state?
It states that $E[r(X)] = \sum_{\text{all } x} r(x) f_X(x)$.
This allows us to find the expected value of a function of X without first finding the pmf of the new transformed variable.

Explain what is meant by the n^{th} moment of X.

The n^{th} moment is defined as $E[X^n]$, where n is a positive integer. Suppose X is a random variable with the following cdf:

$$F_X(x) = P(X \leq x) = \begin{cases} 0 & \text{if } x < 3, \\ 0.12 & \text{if } 3 \leq x < 5, \\ 0.40 & \text{if } 5 \leq x < 6, \\ 0.55 & \text{if } 6 \leq x < 7, \\ 0.72 & \text{if } 7 \leq x < 11, \\ 0.90 & \text{if } 11 \leq x < 12, \\ 1 & \text{if } x \geq 12 \end{cases}$$

Find $E[X^2]$.

We need to first identify the pmf from the jumps in the CDF:
$P(X = 3) = 0.12$
$P(X = 5) = 0.40 - 0.12 = 0.28$
$P(X = 6) = 0.55 - 0.40 = 0.15$
$P(X = 7) = 0.72 - 0.55 = 0.17$
$P(X = 11) = 0.90 - 0.72 = 0.18$
$P(X = 12) = 1 - 0.90 = 0.10$
Using LOTUS: $E[X^2] = 3^2(0.12) + 5^2(0.28) + 6^2(0.15) + 7^2(0.17) + 11^2(0.18) + 12^2(0.10)$. $E[X^2] = 1.08 + 7.0 + 5.4 + 8.33 + 21.78 + 14.4 = \mathbf{57.99}$.

In this exercise, we will review what is meant by the variance of a random variable. This is Section 4.3 in the notes.

Formally define what we mean by the variance of a random variable X. $Var[X] := E[(X - \mu)^2]$, where $\mu = E[X]$. Can you summarize what the formula above means? (We color coded this in the notes for Section 4.3).
Variance is the average squared distance each value is from its center (mean). When we compute the variance of a random variable, what are the units of the variance?
The units are the square units of the original variable X. Define what we mean by the standard deviation of a random variable X. $SD[X] := \sqrt{Var[X]}$.

Prove for a discrete random variable X, $Var[X] = E\big[X^2\big] - \big(E[X]\big)^2$.
$Var[X] = E[(X-\mu)^2] = E[X^2 - 2\mu X + \mu^2]$.
By linearity of expectation: $E[X^2]-2\mu E[X]+E[\mu^2] = E[X^2]-2\mu(\mu)+\mu^2 = E[X^2] - \mu^2$.

Using the formula in part (e), compute the variance of the random variable Y which has the following probability mass function:

$$f_Y(y) = \begin{cases} 0.5 & \text{if } y = -100, 100 \\ 0 & \text{otherwise} \end{cases}$$

$\mu = E[Y] = -100(0.5) + 100(0.5) = 0$.
$E[Y^2] = (-100)^2(0.5) + (100)^2(0.5) = 10,000(0.5) + 10,000(0.5) = 10,000$.
$Var[Y] = 10,000 - 0^2 = \mathbf{10,000}$.

In this problem, we will review the material of Section 4.4 which covers the Bernoulli and Binomial random variables.

What is the point of a random variable?
To model a process or real-life phenomenon to help predict or forecast quantities of interest. Define the probability mass function of a Bernoulli random variable.
$f_X(x) = p$ if $x = 1$;
$1 - p$ if $x = 0$;
0 otherwise. What does a Bernoulli random variable model? The number of successes in one and only one trial where the outcome is binary (success/failure). Let X model the number of successes when we conduct an experiment with one and only one trial. If this trial results in either a success with probability p or failure with probability $1-p$, then prove that $X \sim Bern(p)$. $P(X = 1) = P(\text{success}) = p$ and $P(X = 0) = P(\text{failure}) = 1 - p$. This matches the Bernoulli definition. Give an example of an experiment that can be modeled by a Bernoulli random variable. Flipping a fair coin once where $X = 1$ if heads and $X = 0$ if tails. What is the support of the Bernoulli random variable? $S_x = \{0, 1\}$

If $X \sim Bern(p)$, then sketch the cumulative distribution function of X.

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If $X \sim Bern(p)$ then prove that $E[X] = p$.

$E[X] = 0(1-p) + 1(p) = p$.

If $X \sim Bern(p)$ then prove that $Var[X] = p(1-p)$.

$E[X^2] = 0^2(1-p) + 1^2(p) = p$.
$Var[X] = E[X^2] - (E[X])^2 = p - p^2 = p(1-p)$. Define the probability mass function of a Binomial random variable.

$f_X(x) = \binom{n}{x} p^x (1-p)^{n-x}$ for $x = 0, 1, \ldots, n$.

What does a Binomial random variable model?

The total number of successes in n independent trials.

Give an example of an experiment that can be modeled by a Binomial random variable.

Rolling a die 2025 times and counting how many times a "5" appears.

What is the support of the Binomial random variable?

$S_x = \{0, 1, 2, \ldots, n\}$.

If $X \sim Binomial(3, 0.6)$, then sketch the cumulative distribution function of X.

A step function with jumps at each integer from 0 to n.

If $X \sim Binom(n, p)$ then what is $E[X]$? You do not have to prove your answer. What is $Var[X]$? Again you do not have to prove your answer, just state the results.

$E[X] = np$ and $Var[X] = np(1 - p)$.

On a multiple-choice exam with 3 possible answers for each of the 5 questions, what is the probability that a student will get 4 or more correct answers just by guessing? (You may assume that the questions are independent of each other). You may leave your answer in terms of a calculator command.
$P(X \geq 4) = P(X = 4) + P(X = 5)$. $n = 5, p = 1/3$
Command: 1 - binomcdf(5, 1/3, 3)

Suppose the probability that a certain experiment will be successful is 0.4 and let X denote the number of successes that are obtained in 15 independent performances of the experiment. Find $P(6 \leq X \leq 9)$. You may leave your answer in terms of a calculator command.
$P(6 \leq X \leq 9)$ with $n = 15, p = 0.4$:
Command: binomcdf(15, 0.4, 9) - binomcdf(15, 0.4, 5).

A certain electronic system contains 10 components. Suppose that the probability that each individual component will fail is 0.2 and that the components fail independently of each other. Given that at least one of the components has failed, what is the probability that at least two of the components have failed? You may leave your answer in terms of a calculator command.
$P(X \geq 2 | X \geq 1) = \frac{P(X \geq 2)}{P(X \geq 1)} = \frac{1 - P(X \leq 1)}{1 - P(X=0)}$.
Command: (1 - binomcdf(10, 0.2, 1)) / (1 - binompdf(10, 0.2, 0)).

In this problem we will review the random variable(s) covered in Section 4.5.

What is the pmf of the Geometric random variable? $f(x) = (1-p)^{x-1}p$ for $x = 1, 2, 3, \ldots$. What is the support of the Geometric random variable? $S_x = \{1, 2, 3, \ldots\}$ Under what conditions do we obtain a Geometric random variable? Repeating independent trials with constant probability p until the first success occurs, then stopping. Give an example of an experiment that can be modeled by a Geometric random variable. Tossing a coin repeatedly until the first "Heads" appears. What is the cdf of the Geometric random variable? There is no need to prove your answer, just state the result. $P(X \leq k) = 1 - (1-p)^k$ Suppose $X \sim Geo(p)$. Show that $P(X > m+n \mid X > n) = P(X > m)$. This is what is known as the *memorylessness* property. You may assume that m and n are in the support of X. $P(X > m + n | X > n) = \frac{P(X > m+n)}{P(X>n)} = \frac{(1-p)^{m+n}}{(1-p)^n} = (1-p)^m = P(X > m)$. At Coney Island, you

can play a game that requires the player to shoot a basketball seven times. If a player makes at least six of the seven shots, then they win a prize and the game is over. If a player does not make at least six of the seven shots, then they lose and the game is over. A contestant, Stephen Curry, continually plays the game until he is able to win a prize at which point he stops. Find the probability that Curry will have to play the game more than three times. You may assume that the probability Curry makes any shot is 82% and each shot is independent of the others. You may also assume that the games are independent of each other. You may leave your answer in terms of a calculator command. First, find the probability of winning one game
(p_{win}): $p_{win} = P(\text{make } \geq 6 \text{ of } 7) = \text{binompdf}(7, 0.82, 6) + \text{binompdf}(7, 0.82, 7)$.
Let Y be the number of games played until a win. $Y \sim Geo(p_{win})$.
We want $P(Y > 3) = (1 - p_{win})^3$.
$Command : (1 - (binompdf(7, .82, 6) + binompdf(7, .82, 7)))^3$ A student wishes to take a driving test in order to obtain their learners permit. There are exactly 20 questions on the New York permit driving test. All of the questions on the test are multiple choice, with four possible answers. In order to pass the permit test, the student has to correctly answer **at least** 14 of the 20 questions. However, the student is unprepared and comes up with the following strategy. They decide they will not study and randomly guess the answer to each question. They will then repeatedly take the test until they are able to pass the exam at which point they will stop. Find the probability that the student will have to take the test more than fifty times. (Note: Each question on the test has only one correct answer. You may assume each question is independent of other questions, and each time they take an exam is independent of other exams). $p_{pass} = P(X \geq 14)$ where $X \sim Binom(20, 0.25)$.
Let $Y \sim Geo(p_{pass})$. We want $P(Y > 50) = (1 - p_{pass})^{50}$.
$Command : (binomcdf(20, 0.25, 13))^50$.